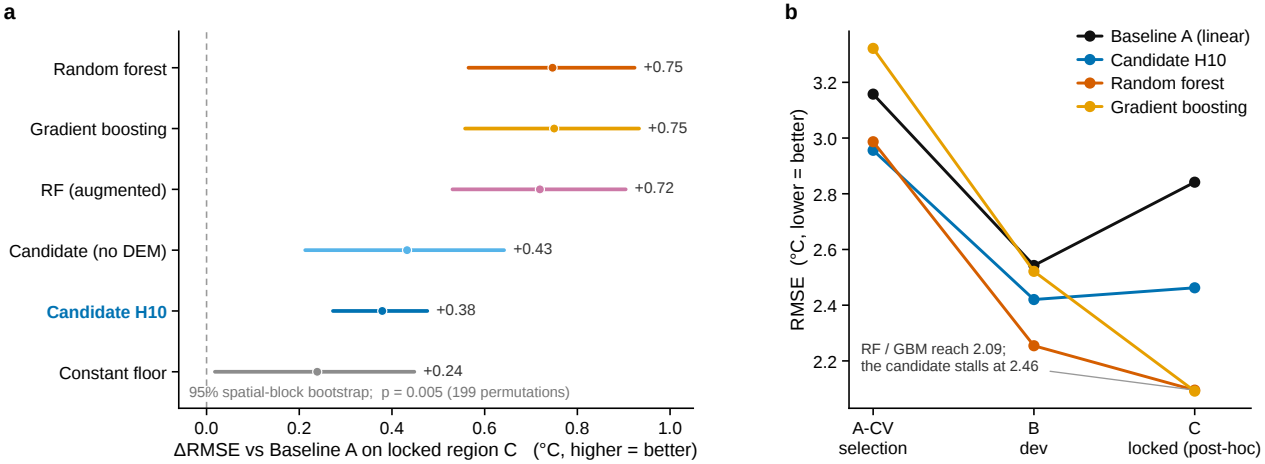
a
Random forest
Gradient boosting
RF (augmented)
Candidate (no DEM)
Candidate H10
Constant floor
+0.75
+0.75
+0.72
+0.43
+0.38
+0.24
95% spatial-block bootstrap; p = 0.005 (199 permutations)
0.0
0.2
0.4
0.6
0.8
1.0
ΔRMSE vs Baseline A on locked region C (°C, higher = better)
b
Baseline A (linear)
Candidate H10
Random forest
Gradient boosting
RMSE (°C, lower = better)
3.2
3.0
2.8
2.6
2.4
2.2
RF / GBM reach 2.09;
the candidate stalls at 2.46
A-CV
selection
B
dev
C
locked (post-hoc)